MIL-C-60103(MU)

30 JANUARY 1964

SUPERSEDING
FA-PD-MI-2392, REV. 2
12 FEBRUARY 1963

MILITARY SPECIFICATION

CUTTER, POWDER ACTUATED, M8 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification is the applicable detail specification for Cutter, Powder Actuated, M8, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

FEDERAL

FED-STD-141 — Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-413 — Visual Inspection Guide for Rubber O-Rings.

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel.

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

DRAWINGS

MUNITIONS COMMAND

F10520567 — Cutter, Powder Actuated, M8, Metal Parts Assembly

C8835919 — Container, Ammunition, Fiber M436 for Cutter, Cartridge Actuated, M8 with Cartridge M122.

C8835920 — Box, Packing, Ammunition for Cutter, Cartridge Actuated, M8 with Cartridge M122 in Fiber Container M436.

IEL–10520567 — Index of Inspection Equipment List for Cutter, Powder Actuated, M8, Metal Parts Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

D149–61 — Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial sample shall be required.

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties. Physical properties required by the drawing or specification shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure. The body and swivel shall withstand a constant internal pressure of 1500 pounds per square inch (psi), minimum, for 15 seconds, minimum, without application of any external load other than that applied by the liquid pressure and shall show no evidence of leakage or mechanical failure (see 6.1).

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.6.1 *Torque.* Torque requirements shall be as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE I

Component	*A* Run on, minimum (inch-pounds)	*B* Seating or bottoming (including A) (inch-pounds)
Cap and swivel	40	300 + 15
Head and body	40	(*)

* Not required. Head shall be run on body to a position where initial bottom contact is made and then no further torquing need be performed.

3.6.2 *Swivel rotation.* Upon completion of assembly, the swivel shall be free to rotate 360 degrees.

3.7 **Protective finish.** Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.8 **Blade coating.** The requirements for blade coating are as specified on the applicable drawing and the following.

3.8.1 *Electrical insulation.* The coating shall be capable of withstanding 1000 volts AC without suffering puncture, charring or any other physical damage.

3.8.2 *Solvent resistance.* The coating shall show no evidence of peeling, blistering, dissolving or loosening when subjected to acetone exposure.

3.8.3 *Appearance.* The coating shall be free from blisters, cracks, wrinkles or other irregularities.

3.9 **Residual magnetism.** The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.10 **Workmanship.** The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.10.2 *Bur.* No part shall have a bur which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Initial production sample.** Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evalution. The sample shall consist of 5 assemblies, 10 sets of assembled parts and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

50 shear pins for the same lot of each type of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with Standard MIL–STD–105.

4.3.1.2 *Size of lot.* The shipping lot of metal parts assemblies shall consist of 318 cutters, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and criteria in tables II and III. For an individual defect (single characteristic) the criteria in table II shall be used. For total defects within a major or minor group the criteria in table III shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Inspection for O-rings shall be in accordance with Standard MIL–STD–413. Four power (4X) magnification may be used in performing visual examination for such defects as bur and foreign matter.

TABLE II

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 — 300	100% Examination				
301 — 1,300	110	1	2	2	3
1,301 — 3,200	150	2	3	3	4
3,201 — 8,000	225	3	4	4	5
8,001 — 22,000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

TABLE III. *Total allowable defects per sample size*

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class:				
2 – 3	1	2	3	4
4 – 7	2	3	4	5
8 – 19	3	4	5	7
20 – 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class:				
2 – 3	2	3	4	5
4 – 7	3	4	5	7
8 – 16	4	5	8	10
17 – 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 *Pin, Shear, Dwg. B8593876, covering a detail of Dwg. F10520567.*

Defect	*Method of inspection*
Critical:	
None defined	
Major:	
101. Diameter, maximum	Gage
102. Total length, minimum	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Bur	Visual
106. Foreign matter	Visual
Minor:	
201. Total length, maximum	Gage
202. Profile of ends	Visual
203. Surface finish, improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.2 *Body Assemb'y, Dwg. F10520569, covering a detail of Dwg. F10520567 (prior to inserting pellets).*

Defect	*Method of inspection*
Critical:	
None defined	
Major:	
101. Pitch diameter of long thread, minimum	Gage
102. Major diameter of long thread, minimum	Gage
103. Pitch diameter of short thread, minimum	Gage
104. Major diameter of short thread, minimum	Gage
105. Concentricity of datum diameter with large bore	Gage
106. Concentricity of small bore with large bore	Gage
107. Concentricity of counterbore with large bore	Gage
108. Concentricity of large bore with pitch diameter of short thread	Gage
109. Diameter of large bore	Gage
110. Diameter of counterbore	Gage
111. Location of datum diameter	Gage
112. Thread damaged or not full	Visual
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish of large and small bores, improper	Visual
116. Bur or sharp corner	Visual
117. Foreign matter	Visual
Minor:	
201. Pitch diameter of long thread, maximum	Gage
202. Pitch diameter of short thread, maximum	Gage
203. Diameter of small bore	Gage
204. Diameter of body	Gage
205. Depth of counterbore	Gage
206. Total length	Gage
207. Diameter of undercut, long thread, minimum	Measure
208. Diameter of undercut, short thread, minimum	Measure
209. Width of undercut, long thread	Measure
210. Width of undercut, short thread	Measure
211. Location of undercut, long thread	Measure
212. Location of undercut, short thread	Measure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect		*Method of inspection*
213.	Radius of chamfer missing or incorrect	Visual
214.	Surface finish improper, except as otherwise classified	Visual
215.	Marking missing, incorrect, or illegible	Visual

4.3.2.3 *Body Assembly, Dwg. F10520569, covering a detail of Dwg. F10520567 (after inserting pellets).*

Critical:

None defined

Major:

101.	Protrusion of pellets above major diameter of thread	Gage
102.	Pellet missing, loose or improperly inserted	Manual/Visual
103.	Thread damaged	Visual
104.	Bur	Visual
105.	Foreign matter	Visual

Minor:

201.	Location of pellets	Gage

4.3.2.4 *Blade, Dwg. D10520572, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101.	Concentricity of body with blade	Gage
102.	Concentricity of groove with body	Gage
103.	Concentricity of pin holes	Gage
104.	True position of pin holes	Gage
105.	Diameter of body	Gage
106.	Diameter of groove	Gage
107.	Diameter of blade	Gage
108.	Diameter of pin holes, miximum	Gage
109.	Profile of groove	Measure
110.	Profile of cutting edge	Measure
111.	Cutting edge, flat	Visual
112.	Metal defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
113. Protective finish inadequate or defective	Visual
114. Surface preparation, improper or inadequate	Visual
115. Surface finish of blade, improper	Visual
116. Bur or sharp edges, except as otherwise classified	Visual
117. Foreign matter	Visual
Minor:	
201. Diameter of pin holes, minimum	Gage
202. Diameter of cavity	Gage
203. Depth of cavity	Gage
204. Length of body	Gage
205. Total length	Gage
206. Length of angle on cutting edge	Gage
207. Width across base of cutting edge	Gage
208. Location of pin holes	Gage
209. Location of groove	Gage
210. Width of groove	Measure
211. Thickness of blade	Measure
212. Radius of chamfer missing or incorrect	Visual
213. Coating, improper, inadequate or defective	Visual
214. Surface finish improper, except as otherwise classified	Visual
215. Marking missing, incorrect or illegible	Visual

4.3.2.5 *Head, Dwg. D10520573, covering a detail of Dwg. F10520567.*

Critical:	
None defined	
Major:	
101. Pitch diameter of large internal thread, maximum	Gage
102. Minor diameter of large internal thread, maximum	Gage
103. Pitch diameter of gas port thread, maximum	Gage
104. Minor diameter of gas port thread, maximum	Gage
105. Concentricity of long cavity with pitch diameter of large thread	Gage
106. Concentricity of boss with pitch diameter of large thread	Gage
107. Perpendicularity of boss face with long cavity	Gage
108. True position of pin hole	Gage
109. Diameter of boss, maximum	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	*Method of inspection*
110. Diameter of long cavity	Gage
111. Diameter of pin hole	Gage
112. Depth of counterbore at mouth of gas port	Gage
113. Length of thread, gas port, minimum	Gage
114. Length of effective thread, set screw, minimum	Gage
115. Location of pin hole	Gage
116. Height of boss	Gage
117. Diameter of chamfer, gas port	Measure
118. Angle of chamfer, gas port	Measure
119. Thread damaged or not full	Visual
120. Metal defective	Visual
121. Protective finish inadequate or defective	Visual
122. Surface finish of long cavity and angle of gas port, improper	Visual
123. Bur or sharp corner	Visual
124. Foreign matter	Visual
Minor:	
201. Pitch diameter of large internal thread, minimum	Gage
202. Pitch diameter of gas port thread, minimum	Gage
203. Pitch diameter of set screw thread	Gage
204. Minor diameter of set screw thread, maximum	Gage
205. Diameter of boss, minimum	Gage
206. Diameter of counterbore, gas port	Gage
207. Diameter of body, large end	Gage
208. Diameter of hole in gas port	Gage
209. Depth of large thread cavity	Gage
210. Depth of set screw cavity, maximum	Gage
211. Length of body, large end	Gage
212. Total length	Gage
213. Height of gas port	Gage
214. Location of gas port cavity	Gage
215. Diameter of body, small end	Measure
216. Diameter of undercut, large thread, maximum	Measure
217. Depth of gas port cavity	Measure
218. Depth to drill point, maximum	Measure
219. Depth to bottom of long cavity, minimum	Measure
220. Width of undercut, large thread	Measure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect		Method of inspection
221.	Profile of gas port	Measure
222.	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
223.	Surface finish improper, except as otherwise classified	Visual
224.	Marking missing, incorrect or illegible	Visual

4.3.2.6 *Pin, Firing, Dwg. C10520574, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101.	Concentricity of striker with body	Gage
102.	Concentricity of groove with body	Gage
103.	True position of pin hole with body	Gage
104.	Diameter of groove	Gage
105.	Diameter of pin hole, maximum	Gage
106.	Depth of pin hole, maximum	Gage
107.	Length of striker	Gage
108.	Profile of striker	Measure
109.	Profile of groove	Measure
110.	Metal defective	Visual
111.	Protective finish inadequate or defective	Visual
112.	Bur or sharp corner	Visual
113.	Surface finish, improper	Visual
114.	Foreign matter	Visual

Minor:

201.	True position of wrench holes with body	Gage
202.	Diameter of body	Gage
203.	Diameter of pin hole, minimum	Gage
204.	Depth of pin hole, minimum	Gage
205.	Total length	Gage
206.	Width across wrench holes	Gage
207.	Location of pin hole	Gage
208.	Diameter of wrench holes	Measure
209.	Depth of wrench holes	Measure
210.	Width of groove	Measure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
211. Location of groove	Measure
212. Radius missing or incorrect	Visual
213. Marking missing, incorrect, or illegible	Visual

4.3.2.7 *Anvil, Dwg. B10520575, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Concentricity of pilot with body	Gage
102. Metal defective	Visual

Minor:

201. Diameter of body	Gage
202. Diameter of hole	Gage
203. Diameter of pilot	Gage
204. Length of body	Gage
205. Total length	Gage
206. Location of hole	Gage
207. Radius or chamfer missing or incorrect	Visual
208. Protective finish inadequate or defective	Visual
209. Bur or sharp edge	Visual
210. Surface finish, improper	Visual
211. Marking missing, incorrect, or illegible	Visual
212. Foreign matter	Visual

4.3.2.8 *Clip, Cable Holder, Dwg. B10520576, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Metal defective	Visual

Minor:

201. Diameter of hole	Gage
202. Total length	Measure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
203. Width of gap	Measure
204. Height	Measure
205. Location of hole	Measure
206. Protective finish inadequate or defective	Visual
207. Surface finish, improper	Visual
208. Marking missing, incorrect, or illegible	Visual
209. Bur	Visual
210. Foreign matter	Visual

4.3.2.9 *Pin, Shear, Unlock, Dwg. B10520577, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Diameter, maximum	Gage
102. Total length, minimum	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Bur or sharp edges	Visual
106. Foreign matter	Visual

Minor:

201. Total length, maximum	Gage
202. Surface finish, improper	Visual

4.3.2.10 *Cap, Dwg. C10520578, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Concentricity of hole with pitch diameter of thread	Gage
102. Pitch diameter of thread, maximum	Gage
103. Minor diameter of thread, maximum	Gage
104. Thread damaged or not full	Visual
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	*Method of inspection*
107. Bur or sharp edges	Visual
108. Foreign matter	Visual
Minor:	
201. Pitch diameter of thread, minimum	Gage
202. Diameter of hole	Gage
203. Depth of cavity	Gage
204. Diameter of undercut, maximum	Measure
205. Total length	Measure
206. Width of undercut	Measure
207. Width across flats	Measure
208. Radius or chamfer missing or incorrect	Visual
209. Surface finish, improper	Visual
210. Marking missing, incorrect, or illegible	Visual

4.3.2.11 *Nut, Mounting, Dwg. B10520579, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Pitch diameter of thread, maximum	Gage
102. Minor diameter of thread, maximum	Gage
103. Thread damaged or not full	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Bur or sharp edges	Visual
107. Foreign matter	Visual
Minor:	
201. Pitch diameter of thread, minimum	Gage
202. Diameter of drill holes	Measure
203. Diameter across corners	Measure
204. Width across flats	Measure
205. Thickness	Measure
206. Location of drill holes	Visual
207. Drill hole, missing	Visual
208. Chamfer missing or incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
209. Surface finish, improper	Visual
210. Marking missing, incorrect, or illegible	Visual

4.3.2.12 *Gasket, Dwg. B10520580, covering a detail of Dwg. F10520567.*

Critical:

None defined

Major:

101. Outside diameter	Gage
102. Inside diameter	Gage
103. Thickness	Measure
104. Material defective	Visual
105. Foreign matter	Visual

Minor:

None defined

4.3.2.13 *Swivel, Assembly, Dwg. F10520581, covering a detail of Dwg. F10520567 (prior to inserting pellets).*

Critical:

None defined

Major:

101. Pitch diameter of thread, minimum	Gage
102. Major diameter of thread, minimum	Gage
103. Concentricity of body with large bore and neck diameter	Gage
104. Concentricity of small bore with large bore	Gage
105. Concentricity of internal datum with large bore	Gage
106. Concentricity of external datum with neck diameter	Gage
107. Concentricity of pin holes	Gage
108. Concentricity of counterbore with large bore	Gage
109. Concentricity of small bore with pitch diameter of thread	Gage
110. Symmetry of external flats with neck	Gage
111. Symmetry of internal flats with small bore	Gage
112. True position of pin holes	Gage
113. Diameter of large bore, maximum	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

	Defect	*Method of inspection*
114.	Diameter of counterbore, maximum	Gage
115.	Diameter of body	Gage
116.	Diameter of pin holes, maximum	Gage
117.	Depth of counterbore	Gage
118.	Length to datum diameter, external	Gage
119.	Length of datum diameter, internal	Gage
120.	Location of pin holes	Gage
121.	Thread damaged or not full	Visual
122.	Metal defective	Visual
123.	Protective finish inadequate or defective	Visual
124.	Surface finish of body, large bore and angles improper	Visual
125.	Bur or sharp corners	Visual
126.	Foreign matter	Visual

Minor:

201.	Pitch diameter of thread, maximum	Gage
202.	Diameter of large bore, minimum	Gage
203.	Diameter of counterbore, minimum	Gage
204.	Diameter of pin holes, minimum	Gage
205.	Diameter of body relief	Gage
206.	Diameter of neck	Gage
207.	Diameter of small bore	Gage
208.	Length of slot through neck	Gage
209.	Total length	Gage
210.	Width of slot through neck	Gage
211.	Diameter of thread undercut, minimum	Measure
212.	Length of flat on neck	Measure
213.	Length to body relief	Measure
214.	Length to end of body relief	Measure
215.	Width of thread undercut	Measure
216.	Width across flat on neck	Measure
217.	Location of thread undercut	Measure
218.	Location of flat on neck	Measure
219.	Radius or chamfer missing or incorrect	Visual
220.	Surface finish improper, except as otherwise classified	Visual
221.	Marking missing, incorrect, or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.14 *Swivel Assembly, Dwg. F10520581, covering a detail of Dwg. F10520567 (After inserting pellets).*

Defect	*Method of inspection*
Critical:	
None defined	
Major:	
101. Protrusion of pellets, maximum	Gage
102. Pellet missing, loose or improperly inserted	Manual/Visual
103. Thread damaged	Visual
104. Bur	Visual
105. Foreign matter	Visual
Minor:	
201. Location of pellets	Gage

4.3.2.15 *Cutter, Powder Actuated, M8, Metal Parts Assembly, Dwg. F10520567.*

Critical:	
None defined	
Major:	
101. Swivel not free to rotate 360 degrees	Manual
102. Blade shear pin, missing	Visual
103. Any part missing except as otherwise classified	Visual
104. Improper assembly	Visual
105. Index hole in cap, missing	Visual
106. Lubricant missing or excessive	Visual
107. Marking for matching parts identification missing, incorrect or illegible	Visual
108. Bur	Visual
109. Foreign matter	Visual
Minor:	
201. Barrier material in set screw, missing	Visual
202. Shipping plug missing	Visual
203. Locking wire missing	Visual
204. Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.16 *Fiber Container, Dwg. C8835919.*

Defect	*Method of inspection*
Critical:	
None defined	
Major:	
None defined	
Minor:	
201. Gap between body and cover, maximum	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.17 *Packing Box, Dwg. C8835920.*

Critical:	
None defined	
Major:	
None defined	
Minor:	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4 except as provided in Specification MIL–I–45208.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each body and each swivel shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.3.3 *Shear pin strength.* Fifty pins of each type from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of the drawing, as applicable, shall be cause for rejection of the lot.

4.3.3.4 *Torque.* Each cap and swivel subassembly shall be subjected to the torque application specified in table I. Failure of any subassembly to produce the requirements specified shall be cause for rejection of the subassembly.

4.3.3.5 *Protective finish.* The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.6 *Blade coating.*

4.3.3.6.1 Eight blades from each lot of blades shall be tested for thickness, electrical insulation and solvent resistance. Failure of any unit to comply with the requirements of the drawing and 3.8.1 through 3.8.3 as applicable shall be cause for rejection of the lot.

4.3.3.7 *Residual magnetism.* Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.8 shall be cause for rejection of the assembly.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL 10520567, except as provided in Specification MIL–I–45208.

4.4 **Test methods and procedures.**

4.4.1 *Hardness.* Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawing.

4.4.2 *Hydrostatic pressure.* The test shall be performed prior to application of protective finish. The body and swivel, individually, shall be mounted in a device and subjected to the hydrostatic test in accordance with the requirements of 3.5.

4.4.3 *Shear pin strength.* After application of protective finish, a load shall be applied at the steady rate of 0.125 inch per minute.

4.4.4 *Torque.* The torque shall be applied to the cap prior to drilling of hole in the cap and assembly of the lock wire.

4.4.5 *Protective finish.* Test procedure for protective finish shall be in accordance with the applicable military standard referenced on the drawing.

4.4.6 *Blade coating.*

4.4.6.1 *Thickness.* Test procedures shall be performed in accordance with Federal Test Standard Number 141, Method 6181 except that the blades shall be used in lieu of test specimens.

4.4.6.2 *Electrical insulation.* The electrical insulation test shall be performed in accordance with ASTM Standard — D149–61, except that an electrode of ¼ inch diameter rounded to a ⅛-inch radius shall be used, AC voltage to be impressed at the rate of 250 to 300 volts per second until 1000 volts is reached at which time the electrode is immediately removed from the coating. The coating is then visually examined to determine compliance with the requirements of 3.8.1.

4.4.6.3 *Solvent resistance.* The coated part shall be rubbed lightly by hand for 30 seconds with a clean white cloth that has been saturated with acetone. Upon completing the test, the coating shall be air dried and visually examined to determine compliance with the requirements of 3.8.2.

4.4.7 *Residual magnetism.* The cutter shall be tested for excessive residual mag-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

netism in an area free of local magnetic effects by placing the cutter 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The cutter shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the cutter.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Level A — (World Wide Shipment) — Packaging and marking shall be in accordance with Dwgs. C8835919 and C8835920.

5.2 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375–A608

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.